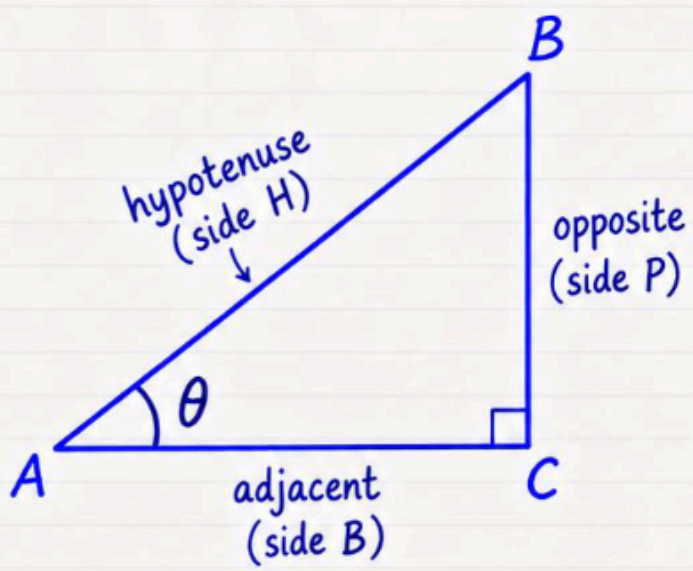
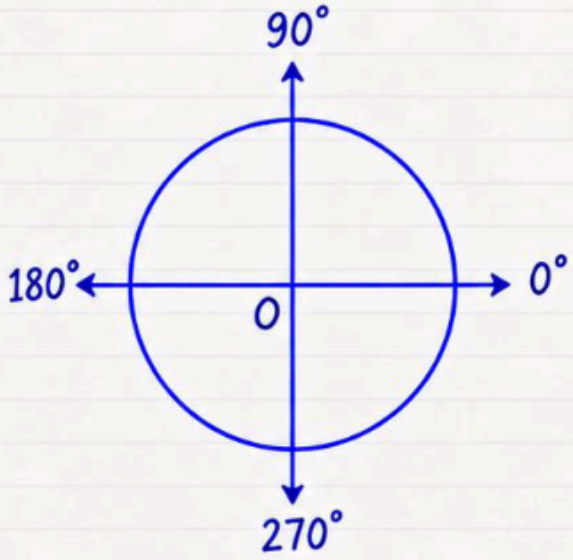
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# TRIGONOMETRY

BASICS • FORMULAS • EXAMPLES • SOLUTIONS




$$\sin\theta = \frac{P}{H} \quad | \quad \cos\theta = \frac{B}{H} \quad | \quad \tan\theta = \frac{P}{B}$$

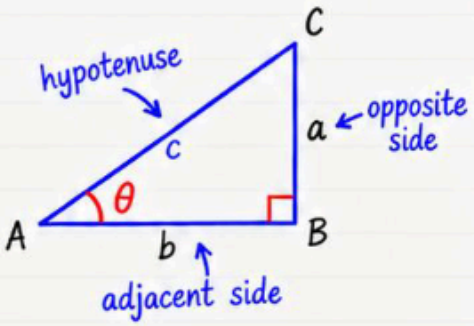
15 - PAGE REVISION NOTES

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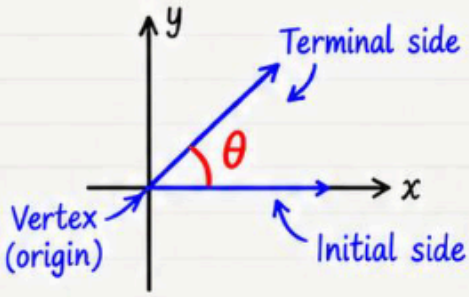
# BASICS & ANGLE MEASUREMENT

## ① What is Trigonometry?

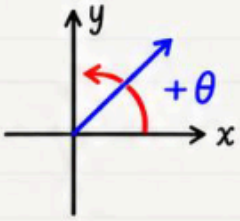
Trigonometry studies the relationships between angles and the sides of triangles. It is widely used in surveying, navigation, engineering and astronomy.



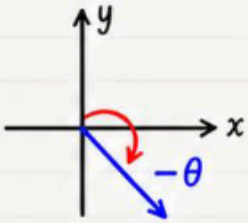

## ② Understanding an Angle




Anticlockwise rotation (Positive angle)




Clockwise rotation (Negative angle)




## ③ Types of Angles

- Acute angle : $0° < \theta < 90°$
- Right angle : $\theta = 90°$
- Obtuse angle : $90° < \theta < 180°$
- Straight angle : $\theta = 180°$
- Reflex angle : $180° < \theta < 360°$
- Complete angle : $\theta = 360°$

## ④ Degrees and Radians

$180° = \pi$ rad
$360° = 2\pi$ rad

Degree to radian:
$$\theta\,(\text{rad}) = \theta\,(°) \times \frac{\pi}{180}$$

Radian to degree:
$$\theta\,(°) = \theta\,(\text{rad}) \times \frac{180}{\pi}$$

**Example 1** : Convert 150° to radians.

$$150° = 150 \times \frac{\pi}{180} = \frac{5\pi}{6} \text{ rad}$$

**Example 2** : Convert $\frac{7\pi}{4}$ radians to degrees.

$$\frac{7\pi}{4} \text{ rad} = \frac{7\pi}{4} \times \frac{180}{\pi} = 315°$$

**Note :** In the arc length formula $s = r\theta$, the angle $\theta$ must be in radians.

@crafted_note

# STANDARD TRIGONOMETRIC VALUES

| $\theta$ | $0^\circ$ | $30^\circ$ | $45^\circ$ | $60^\circ$ | $90^\circ$ |
|---|---|---|---|---|---|
| $\sin\theta$ | $0$ | $\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | $1$ |
| $\cos\theta$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$ | $0$ |
| $\tan\theta$ | $0$ | $\frac{1}{\sqrt{3}}$ | $1$ | $\sqrt{3}$ | undefined |
| $\operatorname{cosec}\theta$ | undefined | $2$ | $\sqrt{2}$ | $\frac{2}{\sqrt{3}}$ | $1$ |
| $\sec\theta$ | $1$ | $\frac{2}{\sqrt{3}}$ | $\sqrt{2}$ | $2$ | undefined |
| $\cot\theta$ | undefined | $\sqrt{3}$ | $1$ | $\frac{1}{\sqrt{3}}$ | $0$ |

## Square-Root Memory Trick

sin values : $\sin 0^\circ = \frac{\sqrt{0}}{2}$, $\sin 30^\circ = \frac{\sqrt{1}}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\sin 90^\circ = \frac{\sqrt{4}}{2}$

cos values : $\cos 0^\circ = \frac{\sqrt{4}}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\cos 45^\circ = \frac{\sqrt{2}}{2}$, $\cos 60^\circ = \frac{\sqrt{1}}{2}$, $\cos 90^\circ = \frac{\sqrt{0}}{2}$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

## Solved Examples

① $2\sin 30^\circ + \cos 60^\circ$

$= 2\left(\frac{1}{2}\right) + \frac{1}{2}$

$= 1 + \frac{1}{2} = \frac{3}{2}$

② $\tan 45^\circ \times \sin 30^\circ$

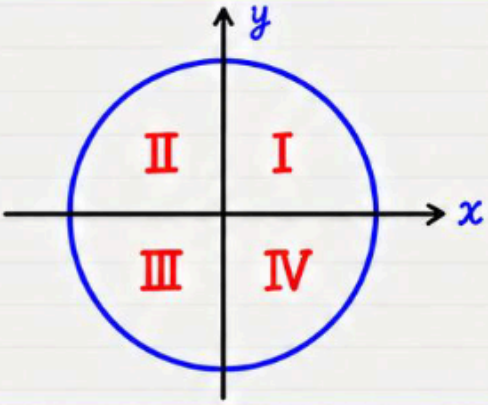
$= 1 \times \frac{1}{2}$

$= \frac{1}{2}$

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# QUADRANTS, SIGNS & REFERENCE ANGLES

## 1. QUADRANTS AND SIGNS




| Quadrant II | Quadrant I |
|---|---|
| sin and cosec positive. | All ratios positive. |
| **Quadrant III** | **Quadrant IV** |
| tan and cot positive. | cos and sec positive. |

**ASTC : All – Sin – Tan – Cos**

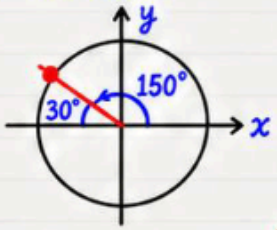
## 2. REFERENCE ANGLE

The reference angle ($\alpha$) is the acute angle between the terminal side and the x-axis.

| QI : | QII : | QIII : | QIV : |
|---|---|---|---|
| $\alpha = \theta$ | $\alpha = 180° - \theta$ | $\alpha = \theta - 180°$ | $\alpha = 360° - \theta$ |

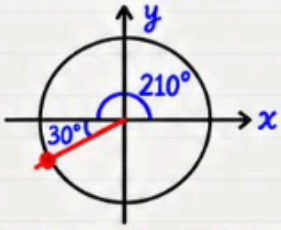
## 3. EXAMPLES

**(i) sin 150°**




$\sin 150° = \sin 30° = \frac{1}{2}$

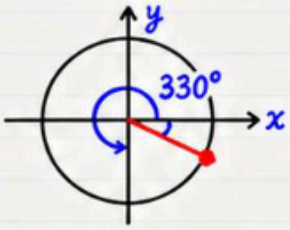
**(ii) cos 210°**




$\cos 210° = -\cos 30°$

$= -\frac{\sqrt{3}}{2}$

**(iii) tan 330°**




$\tan 330° = -\tan 30°$

$= -\frac{1}{\sqrt{3}}$

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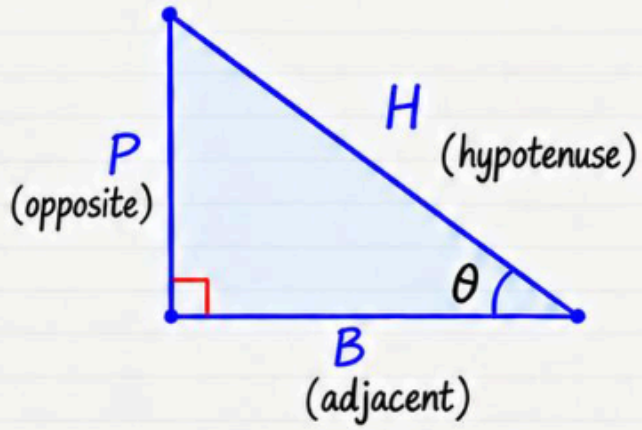
# TRIGONOMETRIC RATIOS

## ① Three Basic Ratios

$$\sin\theta = \frac{P}{H}$$

$$\cos\theta = \frac{B}{H}$$

$$\tan\theta = \frac{P}{B}$$




## ② Reciprocal Ratios

$$\text{cosec}\,\theta = \frac{H}{P} = \frac{1}{\sin\theta} \quad | \quad \sec\theta = \frac{H}{B} = \frac{1}{\cos\theta} \quad | \quad \cot\theta = \frac{B}{P} = \frac{1}{\tan\theta}$$

## ③ Memory Trick

**SOH**

$$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

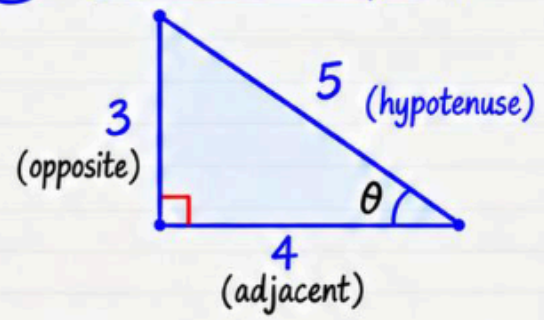
**CAH**

$$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

**TOA**

$$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

## ④ Solved Example (3-4-5 Triangle)




$$\sin\theta = \frac{3}{5} \quad \cos\theta = \frac{4}{5} \quad \tan\theta = \frac{3}{4}$$

$$\text{cosec}\,\theta = \frac{5}{3} \quad \sec\theta = \frac{5}{4} \quad \cot\theta = \frac{4}{3}$$

**Note:** The opposite and adjacent sides depend on the chosen angle $\theta$, but the hypotenuse is always opposite 90°.

@crafted_note

# FUNDAMENTAL IDENTITIES

## ① Reciprocal Identities

$$\sin\theta = \frac{1}{\operatorname{cosec}\theta} \qquad \operatorname{cosec}\theta = \frac{1}{\sin\theta}$$

$$\cos\theta = \frac{1}{\sec\theta} \qquad \sec\theta = \frac{1}{\cos\theta}$$

$$\tan\theta = \frac{1}{\cot\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

## ② Quotient Identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

## ③ Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1 \qquad 1 + \tan^2\theta = \sec^2\theta \qquad 1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

### Example 1 : Simplify

$$\frac{1 - \cos^2\theta}{\sin\theta}$$

$$= \frac{\sin^2\theta}{\sin\theta} \quad (\text{using } \sin^2\theta = 1 - \cos^2\theta)$$

$$= \sin\theta$$

### Example 2 : Prove

$$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$$

$$= \sec^2\theta - \tan^2\theta \quad (\text{using } (a-b)(a+b) = a^2 - b^2)$$

$$= 1 \quad (\text{using } \sec^2\theta - \tan^2\theta = 1)$$

### How to prove identities :

1. Start with the more complicated side.
2. Convert to sin and cos when stuck.
3. Never cancel terms across addition or subtraction.

@crafted_note

# COMPLEMENTARY & ALLIED ANGLES

## 1. COMPLEMENTARY ANGLES

| | |
|---|---|
| $\sin(90^\circ - \theta) = \cos\theta$ | $\cot(90^\circ - \theta) = \tan\theta$ |
| $\cos(90^\circ - \theta) = \sin\theta$ | $\sec(90^\circ - \theta) = \operatorname{cosec}\theta$ |
| $\tan(90^\circ - \theta) = \cot\theta$ | $\operatorname{cosec}(90^\circ - \theta) = \sec\theta$ |

## 2. ALLIED-ANGLE RULE

- Odd multiples of $90^\circ$ change the function ($\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \operatorname{cosec}$).
- Multiples of $180^\circ$ keep the same function.
- Determine the sign from the quadrant.

## 3. QUICK TABLE

| Angle | sin | cos | tan |
|---|---|---|---|
| $90^\circ + \theta$ | $\cos\theta$ | $-\sin\theta$ | $-\cot\theta$ |
| $180^\circ - \theta$ | $\sin\theta$ | $-\cos\theta$ | $-\tan\theta$ |
| $180^\circ + \theta$ | $-\sin\theta$ | $-\cos\theta$ | $\tan\theta$ |
| $270^\circ - \theta$ | $-\cos\theta$ | $-\sin\theta$ | $\cot\theta$ |
| $270^\circ + \theta$ | $-\cos\theta$ | $\sin\theta$ | $-\cot\theta$ |
| $360^\circ - \theta$ | $-\sin\theta$ | $\cos\theta$ | $-\tan\theta$ |
| $360^\circ + \theta$ | $\sin\theta$ | $\cos\theta$ | $\tan\theta$ |

## 4. NEGATIVE & PERIODIC ANGLES

| | |
|---|---|
| $\sin(-\theta) = -\sin\theta$ | $\sin(\theta + 360^\circ) = \sin\theta$ |
| $\cos(-\theta) = \cos\theta$ | $\cos(\theta + 360^\circ) = \cos\theta$ |
| $\tan(-\theta) = -\tan\theta$ | $\tan(\theta + 180^\circ) = \tan\theta$ |

## 5. EXAMPLES

(i) $\cos 60^\circ = \sin 30^\circ = \frac{1}{2}$

(ii) $\sin 135^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$

(iii) $\tan 300^\circ = -\tan 60^\circ = -\sqrt{3}$

@crafted_note

# TRANSFORMATION FORMULAS

## ① Product to Sum

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$
$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$
$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$
$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

Product form

Product to Sum ↕ Sum to Product

Sum form

## ② Sum to Product

$$\sin C + \sin D = 2\sin\left[\frac{C+D}{2}\right]\cos\left[\frac{C-D}{2}\right]$$
$$\sin C - \sin D = 2\cos\left[\frac{C+D}{2}\right]\sin\left[\frac{C-D}{2}\right]$$
$$\cos C + \cos D = 2\cos\left[\frac{C+D}{2}\right]\cos\left[\frac{C-D}{2}\right]$$
$$\cos C - \cos D = -2\sin\left[\frac{C+D}{2}\right]\sin\left[\frac{C-D}{2}\right]$$

**Example 1 : Simplify**

$$\sin 70^\circ + \sin 10^\circ$$
$$= 2\sin\left(\frac{70^\circ+10^\circ}{2}\right)\cos\left(\frac{70^\circ-10^\circ}{2}\right)$$
$$= 2\sin 40^\circ\cos 30^\circ$$
$$= \sqrt{3}\sin 40^\circ$$

**Example 2 : Simplify**

$$2\sin 50^\circ\cos 20^\circ$$
$$= \sin(50^\circ+20^\circ) + \sin(50^\circ-20^\circ)$$
$$= \sin 70^\circ + \sin 30^\circ$$
$$= \sin 70^\circ + \frac{1}{2}$$

@crafted_note

# DOUBLE, TRIPLE & HALF ANGLES

## 1. Double-Angle Formulas

$$\sin 2A = 2\sin A\cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

## 2. Triple-Angle Formulas

$$\sin 3A = 3\sin A - 4\sin^3 A$$

$$\cos 3A = 4\cos^3 A - 3\cos A$$

$$\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

## 3. Half-Angle Formulas

$$\sin\frac{A}{2} = \pm\sqrt{\frac{1-\cos A}{2}}$$

$$\cos\frac{A}{2} = \pm\sqrt{\frac{1+\cos A}{2}}$$

$$\tan\frac{A}{2} = \pm\sqrt{\frac{1-\cos A}{1+\cos A}} = \frac{\sin A}{1+\cos A} = \frac{1-\cos A}{\sin A}$$

**Note:** Choose $\pm$ from the quadrant of $A/2$.

## 4. Examples

**(i) Evaluate $\sin 60°$.**

$$\sin 60° = \sin(2\times 30°) = 2\sin 30°\cos 30° = 2\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{2}$$

**(ii) For $A = 30°$, find $\sin 15°$.**

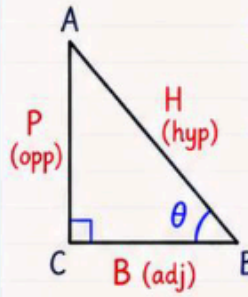
$$\sin 15° = \sin\left(\frac{30°}{2}\right) = \sqrt{\frac{1-\cos 30°}{2}} = \sqrt{\frac{1-\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2-\sqrt{3}}{4}} = \frac{\sqrt{2-\sqrt{3}}}{2}$$

Since $15°$ is in Quadrant I, $\sin 15°$ is positive.

@crafted_note

# MASTER FORMULA & REVISION

## 1. Basic Ratios




$\sin\theta = \frac{P}{H}$ $\quad \cos\theta = \frac{B}{H}$ $\quad \tan\theta = \frac{P}{B}$

$\operatorname{cosec}\theta = \frac{1}{\sin\theta}$ $\quad \sec\theta = \frac{1}{\cos\theta}$ $\quad \cot\theta = \frac{1}{\tan\theta}$ $\quad \tan\theta = \frac{\sin\theta}{\cos\theta}$

## 2. Standard Values

| $\theta$ | $0°$ | $30°$ | $45°$ | $60°$ | $90°$ |
|---|---|---|---|---|---|
| $\sin\theta$ | $0$ | $\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | $1$ |
| $\cos\theta$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$ | $0$ |
| $\tan\theta$ | $0$ | $\frac{1}{\sqrt{3}}$ | $1$ | $\sqrt{3}$ | not defined |

## 3. Core Identities

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

## 4. Angle Formulas

- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- $\sin 2A = 2 \sin A \cos A$
- $\cos 2A = \cos^2 A - \sin^2 A$

## 5. General Solutions

- $\sin x = \sin\alpha \rightarrow x = n\pi + (-1)^n \alpha$
- $\cos x = \cos\alpha \rightarrow x = 2n\pi \pm \alpha$
- $\tan x = \tan\alpha \rightarrow x = n\pi + \alpha$

$(n \in \mathbb{Z})$

## 6. Any Triangle

- Sine Rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
- Cosine Rule: $a^2 = b^2 + c^2 - 2bc\cos A$
- Area of a Triangle: $\text{Area} = \frac{1}{2} bc \sin A$

## 7. Graph Memory

- Period of $\sin\theta = 2\pi$
- Period of $\cos\theta = 2\pi$
- Period of $\tan\theta = \pi$

## 8. Final Checklist

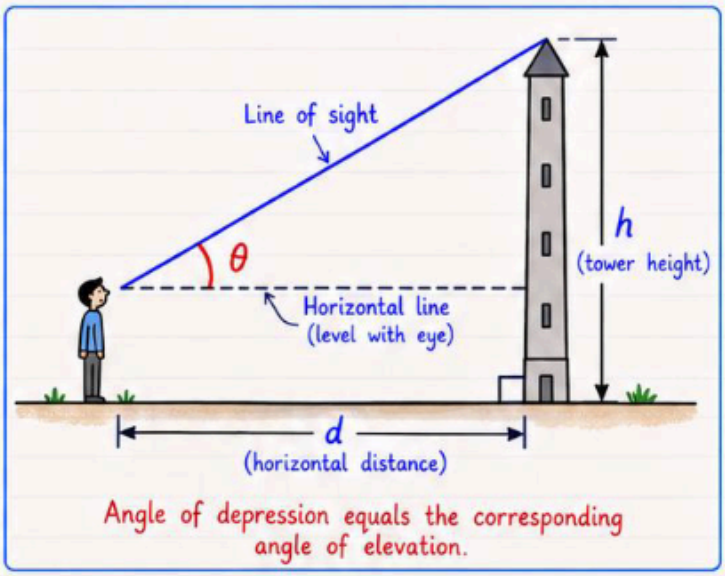
1. Check whether the angle is in degrees or radians.
2. Use the correct quadrant sign for the function.
3. Inverse trigonometric function is not the reciprocal.
4. Denominator in a trigonometric expression cannot be zero.
5. Always obey the given solution interval.

@crafted_note

# HEIGHTS & DISTANCES

## 1. Key Terms

- **Line of sight:** The straight line from the observer's eye to the object being viewed.
- **Angle of elevation:** The angle between the horizontal line and the line of sight when the object is above the observer.
- **Angle of depression:** The angle between the horizontal line and the line of sight when the object is below the observer.
- **Horizontal line:** A line through the observer's eye that is parallel to the ground (level line).




Angle of depression equals the corresponding angle of elevation.

## 2. Steps

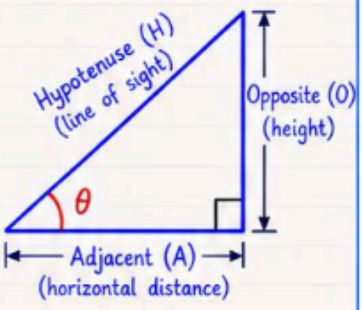
1. Draw the right triangle.
2. Label the known and unknown values (e.g., height $h$, horizontal distance $d$, angle $\theta$).
3. Choose the appropriate trigonometric ratio (sin, cos, or tan) based on the known and unknown sides.
4. Solve for the unknown and write the unit in the final answer.
5. If the observer's height is given, add it to the calculated height of the object.

**Useful Trigonometric Ratios (Right Triangle)**

$$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{O}{H}$$

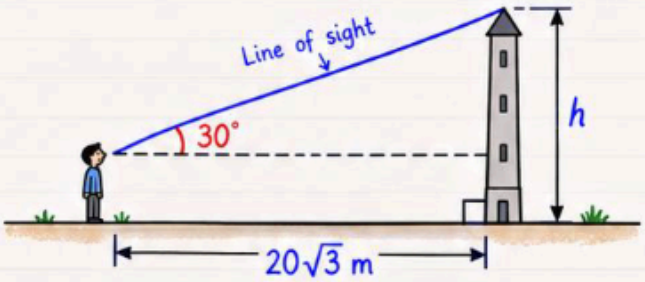
$$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{A}{H}$$

$$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{O}{A}$$




## 3. Worked Example

A tower is viewed from a horizontal distance of $20\sqrt{3}$ m at an elevation angle of 30°. Find the height of the tower.




**Solution:**

$$\tan 30^\circ = \frac{h}{20\sqrt{3}}$$

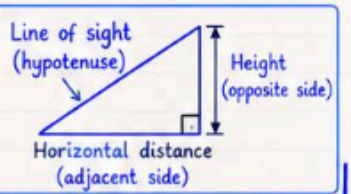
$$\frac{1}{\sqrt{3}} = \frac{h}{20\sqrt{3}} \quad \left(\text{since } \tan 30^\circ = \frac{1}{\sqrt{3}}\right)$$

$$h = 20\sqrt{3} \times \frac{1}{\sqrt{3}}$$

$$h = 20 \text{ m}$$

## 4. Common Mistake

✗ Do not confuse the horizontal distance with the sloping line of sight.

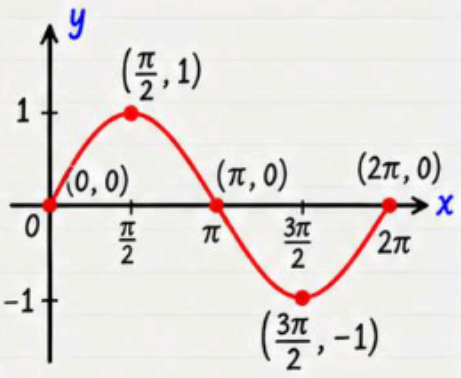


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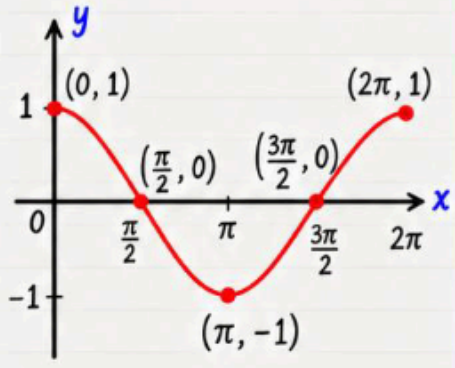
# TRIGONOMETRIC GRAPHS

## 1. GRAPHS OF BASIC TRIGONOMETRIC FUNCTIONS

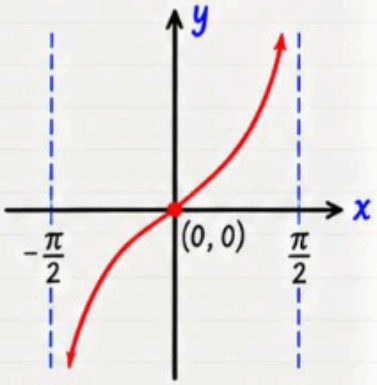
### (i) $y = \sin x$




### (ii) $y = \cos x$




### (iii) $y = \tan x$




## 2. KEY PROPERTIES

| Function | Domain | Range | Period | Amplitude |
|---|---|---|---|---|
| $y = \sin x$ | $\mathbb{R}$ | $[-1, 1]$ | $2\pi$ | 1 |
| $y = \cos x$ | $\mathbb{R}$ | $[-1, 1]$ | $2\pi$ | 1 |
| $y = \tan x$ | $\mathbb{R} \setminus \{\frac{\pi}{2} + n\pi\}$ | $\mathbb{R}$ | $\pi$ | Not defined |

## 3. GRAPH TRANSFORMATION

For $y = A\sin(Bx + C) + D$,

$$\text{Amplitude} = |A|$$

$$\text{Period} = \frac{2\pi}{|B|}$$

$$\text{Phase shift} = -\frac{C}{B}$$

$$\text{Vertical shift} = D$$

**Example :** $y = 2\sin(3x - \pi) + 1$

Amplitude $= |2| = 2$

Period $= \frac{2\pi}{3}$

Phase shift $= -\frac{(-\pi)}{3} = \frac{\pi}{3}$ to the right

Vertical shift $= 1$ (up 1 unit)

Range $= [1-2, 1+2] = [-1, 3]$

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# COMPOUND-ANGLE FORMULAS

## 1. ADDITION & SUBTRACTION

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

$$\sin(A-B) = \sin A\cos B - \cos A\sin B$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$$

**Sign memory:**
- Sine keeps the sign.
- Cosine reverses the sign.
- Tangent follows the numerator sign.

## 2. EXAMPLE 1 : Find sin 75°

$$\begin{aligned}\sin 75^\circ &= \sin(45^\circ + 30^\circ)\\ &= \sin 45^\circ\cos 30^\circ + \cos 45^\circ\sin 30^\circ\\ &= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}\cdot\frac{1}{2}\\ &= \frac{\sqrt{6}+\sqrt{2}}{4}\end{aligned}$$

## 3. EXAMPLE 2 : Find cos 15°

$$\begin{aligned}\cos 15^\circ &= \cos(45^\circ - 30^\circ)\\ &= \cos 45^\circ\cos 30^\circ + \sin 45^\circ\sin 30^\circ\\ &= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}\cdot\frac{1}{2}\\ &= \frac{\sqrt{6}+\sqrt{2}}{4}\end{aligned}$$

## 4. EXAMPLE 3 : Find tan 75°

$$\tan 75^\circ = \tan(45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ\tan 30^\circ} = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$$

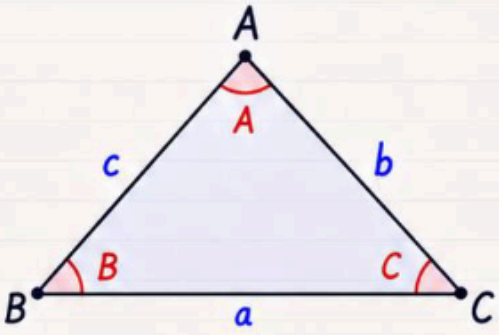
$$= \frac{\sqrt{3}+1/\sqrt{3}}{\sqrt{3}-1/\sqrt{3}} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \frac{(\sqrt{3}+1)^2}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{3+2\sqrt{3}+1}{3-1} = 2+\sqrt{3}$$

@crafted_note

# TRIGONOMETRY IN ANY TRIANGLE

## 1. Triangle and Notation

- Consider a triangle ABC with sides a, b, c opposite to angles A, B, C respectively.
- Side a is opposite angle A, side b is opposite angle B, and side c is opposite angle C.
- Angles A, B, C are usually given in degrees.




## 2. Sine Rule

- In any triangle ABC, the ratios of the sides to the sines of the opposite angles are equal.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

## 3. Cosine Rule

- In any triangle ABC, each side can be expressed in terms of the other two sides and the cosine of the included angle.

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = c^2 + a^2 - 2ca\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

## 4. Area Formula

- The area $\Delta$ of a triangle can be found using two sides and the included angle.

$$\Delta = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \frac{1}{2}ab\sin C$$

## 5. Which Rule?

- Choose the appropriate rule based on the given information.
  - ASA or AAS (two angles and a side) → Use Sine Rule.
  - SSS or SAS (three sides or two sides and included angle) → Use Cosine Rule.
  - Two sides and the included angle → Use Area formula.

## 6. Worked Example

Given: $b = 7$, $c = 5$, $A = 60°$.
Find: (i) side a (ii) area of the triangle.

(i) Find side a (Cosine Rule):

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$= 7^2 + 5^2 - 2(7)(5)\cos 60°$$
$$= 49 + 25 - 70\left(\frac{1}{2}\right)$$
$$= 49 + 25 - 35$$
$$= 39$$
$$a = \sqrt{39} \approx 6.24$$

(ii) Find area (Area formula):

$$\Delta = \frac{1}{2}bc\sin A$$
$$= \frac{1}{2} \times 7 \times 5 \times \sin 60°$$
$$= \frac{35}{2} \times \frac{\sqrt{3}}{2}$$
$$= \frac{35\sqrt{3}}{4} \approx 15.16 \text{ square units}$$

@crafted_note

# INVERSE TRIGONOMETRIC FUNCTIONS

## 1. Meaning

Ans:
- Inverse trigonometric functions (or inverse trig functions) return an angle whose trigonometric value is given.
- If $y = \sin^{-1} x$, then $\sin y = x$ and $y$ is an angle in the principal range.
- $\sin^{-1} x$ is NOT the same as $\frac{1}{\sin x}$.
  $\sin^{-1} x$ means "the angle whose sine is $x$".
- The reciprocal of $\sin x$ is called cosec $x$, i.e.
$$\frac{1}{\sin x} = \text{cosec } x.$$

### Inverse vs Reciprocal (Important Distinction)

| Inverse Function | Reciprocal Function |
|---|---|
| $y = \sin^{-1} x$ | $\frac{1}{\sin x} = \text{cosec } x$ |
| ↓ | ↓ |
| means | means |
| "angle whose sine is $x$" | "reciprocal of $\sin x$" |
| ↓ | ↓ |
| If $y = \sin^{-1} x$, then $\sin y = x$. | It is NOT an angle and is different from $\sin^{-1} x$. |

## 2. Principal Values

Ans:
- The inverse trigonometric functions are multivalued in general.
  We use principal values so that each inverse function returns a unique angle.

| Function | Domain (values of $x$) | Range (principal values) | Remarks |
|---|---|---|---|
| $\sin^{-1} x$ | $[-1, 1]$ | $[-\frac{\pi}{2}, \frac{\pi}{2}]$ | Returns angle $y$ with $\sin y = x$ |
| $\cos^{-1} x$ | $[-1, 1]$ | $[0, \pi]$ | Returns angle $y$ with $\cos y = x$ |
| $\tan^{-1} x$ | $\mathbb{R}$ (all real numbers) | $(-\frac{\pi}{2}, \frac{\pi}{2})$ | Returns angle $y$ with $\tan y = x$ |
| $\cot^{-1} x$ | $\mathbb{R}$ (all real numbers) | $(0, \pi)$ | Returns angle $y$ with $\cot y = x$ (standard school convention) |

## 3. Useful Identities

Ans:
- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
- $\sin(\sin^{-1} x) = x$ (for $-1 \le x \le 1$)
- $\cos(\cos^{-1} x) = x$ (for $-1 \le x \le 1$)

## 4. Examples

Ans:
- $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$
- $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$
- $\tan^{-1}(1) = \frac{\pi}{4}$
- $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \frac{\pi}{4}$

  Since $\sin^{-1}$ returns the principal value in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the angle $\frac{3\pi}{4}$ (which is not in this range) is mapped to $\frac{\pi}{4}$.